

Dip-Coating of a Nanometer-Thick Fluorinated Imidazolium Ionic Liquid: Adsorption Kinetics, Capillarity, and Drainage in Dual-Layer Film Formation

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A study of nanometer-thick film formation is presented for the highly fluorinated imidazolium ionic liquid, HFILOH, deposited on silica via dip-coating. Building on our recently published work [1], which established a dual-layer structure governed by reversible pseudo-first-order adsorption kinetics (bonded layer, $h_{eq} \sim 0.8$ nm) and Landau–Levich concentration-dependent mobile layer entrainment, we now report the complete effect of withdrawal speed at concentrations of 0.1, 0.5, and 1.0 g/L. The bonded layer remains invariant with withdrawal speed at all concentrations, providing direct visual confirmation of its decoupling from hydrodynamics. The mobile layer exhibits two distinct regimes: a capillarity regime at low speeds (≤ 0.5 mm/s) where thickness decreases due to evaporation-driven capillary feeding, and a Landau–Levich drainage regime at high speeds (≥ 1 mm/s) where thickness scales as $U^{2/3}$, amplified at higher concentrations. These findings complete a three-mechanism model for HFILOH nanofilm, providing a complete mechanistic model of HFILOH nanofilm deposition and practical process guidelines for hard disk drive media lubrication and surface engineering applications.

Keywords: adsorption kinetics, capillarity regime, dip-coating, drainage regime, fluorinated ionic liquid, hard disk drive lubrication, Landau–Levich theory.

INTRODUCTION

THE hard disk drive (HDD) industry relies on nanometer-thick lubricant films deposited by dip-coating to protect magnetic media during head-disk contact [2], [3]. For perfluoropolyether (PFPE) lubricants, dip-coating produces a dual-layer film, an adsorption-controlled bonded layer and a hydrodynamically-controlled mobile layer [2], exhibiting capillarity at low withdrawal speeds and Landau–Levich (L–L) drainage at high speeds, with a characteristic minimum near 1 mm/s [3]. The highly fluorinated imidazolium ionic liquid, HFILOH, has been proposed as a next-generation HDD nano-lubricant [4]. In our recently published work [1], we demonstrated that HFILOH forms an analogous dual-layer structure, with the bonded layer governed by reversible pseudo-first-order adsorption kinetics and the mobile layer scaling linearly with concentration via L–L theory at a fixed withdrawal speed. Here we systematically vary withdrawal speed from about 0.02 to 3 mm/s at concentrations of 0.1, 0.5, and 1.0 g/L, providing direct evidence that the bonded layer is invariant with withdrawal speed while the mobile layer exhibits the same capillarity–drainage dual-regime as PFPE lubricants.

EXPERIMENTAL METHODS

HFILOH synthesis and substrate preparation followed [1], [4]. Silicon wafers (2 nm native oxide) were solvent-cleaned and UV/ozone treated for 30 min. Films were deposited from HFILOH/Vertrel XF solutions at 0.1, 0.5, and 1.0 g/L with a fixed dwell time of 5 mins and withdrawal speeds of about 0.02–3 mm/s. Total film thicknesses were measured by spectroscopic ellipsometry (J.A. Woollam Alpha-SE, Cauchy model, $A = 1.45$, $B = 0.0025$); bonded thicknesses after rinsing twice in pure Vertrel XF; mobile as total minus bonded. All values are averages of ≥ 3 independent samples.

RESULTS AND DISCUSSION

A. Bonded Layer: Invariance with Withdrawal Speed

Figures 1–3 show total, bonded, and mobile thicknesses vs. withdrawal speed at 0.1, 0.5, and 1.0 g/L, respectively. Across all concentrations, the bonded layer remains essentially flat: ~ 0.4 nm at 0.1 g/L, ~ 0.6 nm at 0.5 g/L, and ~ 0.6 nm at 1.0 g/L, independent of withdrawal speed from 0.02 to 3 mm/s. This directly confirms that the adsorption mechanism controlling bonded layer formation is fully decoupled from hydrodynamic conditions. Once available surface hydroxyl sites are occupied by HFILOH molecules, no additional bonded material accumulates regardless of how fast the substrate is withdrawn. Table 1 (reproduced from [1]) summarizes the pseudo-first-order kinetic fit parameters: the equilibrium bonded thickness h_{eq} saturates at ~ 0.75 – 0.81 nm across all concentrations, while the time constant τ decreases from ~ 2080 s to ~ 314 s as concentration increases from 0.5 to 1.5 g/L, reflecting faster site saturation at higher concentration.

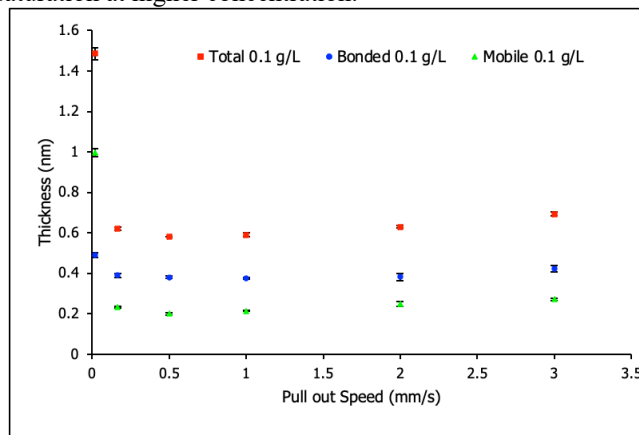


Fig. 1. Total, bonded, and mobile HFILOH thickness vs. withdrawal speed at 0.1 g/L.

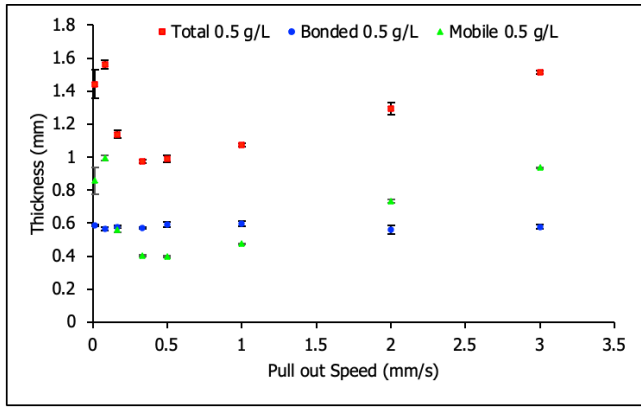


Fig. 2. Total, bonded, and mobile HFILOH thickness vs. withdrawal speed at 0.5 g/L.

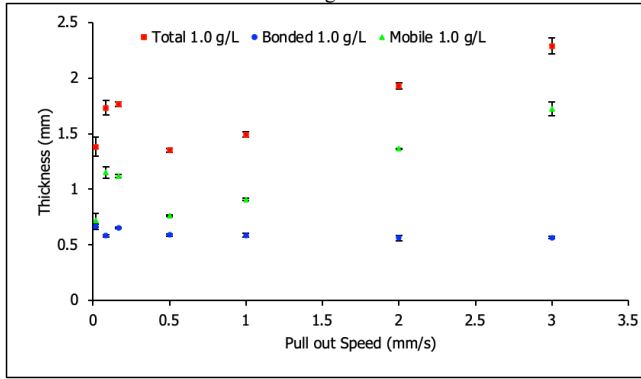


Fig. 3. Total, bonded, and mobile HFILOH thickness vs. withdrawal speed at 1.0 g/L.

Table 1. Pseudo-first-order kinetic fit parameters for bonded layer growth $h_B(t)$ at different concentrations (reproduced from [1]).

Concentration (g/L)	y_0 (nm)	A_0 (nm)	τ (s)	$h_{B,eq}$ (nm)	R^2
0.5	0.445	0.301	2080.701	0.746	0.927
1.0	0.400	0.372	1451.202	0.773	0.984
1.5	0.709	0.097	314.139	0.805	0.757

B. Mobile Layer: Capillarity and Drainage Regimes

Figure 4 overlays the mobile layer thickness vs. withdrawal speed for all three concentrations, revealing two distinct regimes analogous to those reported for PFPE lubricants [3]. In the capillarity regime (≤ 0.5 mm/s), Vertrel XF evaporates faster than the drying line moves; capillary rise feeds additional HFILOH to the surface, and slower withdrawal gives the capillary supply more time to deposit material, so mobile thickness decreases with speed. In the L–L drainage regime (≥ 1 mm/s), viscous entrainment dominates, and mobile thickness increases as $U^{2/3}$ [5]. A minimum near 0.5–1 mm/s marks the crossover. Both regimes are amplified at higher concentrations: at 0.1 g/L, the drainage increase is modest; at 0.5 and 1.0 g/L, both regimes are clearly resolved, consistent with Fan and Waltman’s finding that the mixed regime requires sufficient solute [3].

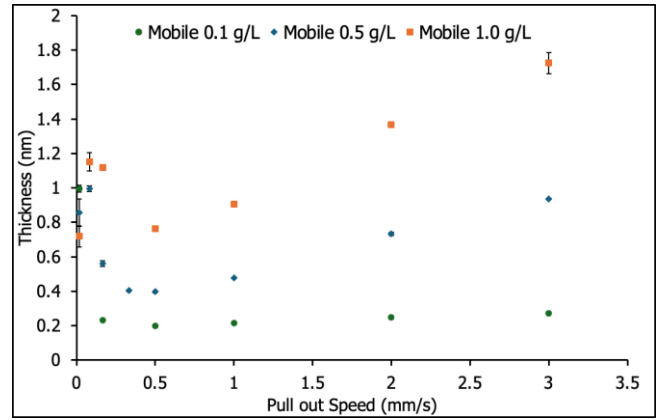


Fig. 4. Mobile layer thickness vs. withdrawal speed at 0.1, 0.5, and 1.0 g/L, showing capillarity (decreasing) and L–L drainage (increasing) regimes with crossover minimum near 0.5–1 mm/s.

C. Unified Three-Mechanism Model

Three decoupled mechanisms govern HFILOH film formation. (1) Adsorption controls the bonded layer: site-limited, speed-independent, saturating at ~ 0.8 nm ($\tau = 314\text{--}2080$ s, Table 1). (2) Capillarity controls the mobile layer at low withdrawal speeds ($h \propto C \cdot U^{-1}$). (3) L–L drainage controls the mobile layer at high speeds ($h \propto C \cdot U^{2/3}$). This provides complete independent control: dwell time and concentration set the bonded layer; concentration and withdrawal speed tune the mobile layer across the $\sim 0.5\text{--}1$ mm/s capillarity–drainage crossover. The typical HDD industry withdrawal speed of 1–2 mm/s places HFILOH deposition in the drainage regime, paralleling PFPE processing [3], while HFILOH’s hydrophilic–oleophobic wetting character (HCA $\sim 70\text{--}75^\circ$, WCA $\sim 30\text{--}40^\circ$ [1]) is preserved across all conditions.

CONCLUSION

A complete three-mechanism model for HFILOH nanofilm formation is established. The bonded layer ($h_{eq} \sim 0.8$ nm, $\tau = 314\text{--}2080$ s) is fully decoupled from withdrawal speed across all concentrations. The mobile layer exhibits capillarity at low speeds and L–L drainage at high speeds, with a crossover minimum near 0.5–1 mm/s amplified by concentration. This model provides practical independent control of both film components, establishing HFILOH as a well-characterized candidate for tribology and head-disk interface applications at >3 Tbits/in².

REFERENCES

- [1] A. Dahiru and L. Li, “Dip-Coating of a Nanometer-Thick Fluorinated Imidazolium Ionic Liquid: Layer Structure, Formation Mechanism and Wetting Behavior,” *Langmuir*, vol. 42, no. 15, pp. 10716–10724, Apr. 2026, doi: 10.1021/acs.langmuir.6c00733.
- [2] A. G. Merzlikine, L. Li, P. M. Jones, and Y.-T. Hsia, “Lubricant layer formation during the dip-coating process: influence of adsorption and viscous flow mechanisms,” *Tribol. Lett.*, vol. 18, no. 3, pp. 279–286, Mar. 2005, doi: 10.1007/s11249-004-2753-8.
- [3] A. Fan and R. J. Waltman, “Dip-Coating perfluoropolyether lubricants on rigid magnetic disks,” *Chem. Data Collect.*, vol. 17–18, pp. 451–458, Dec. 2018, doi: 10.1016/j.cdc.2018.11.007.
- [4] B. Wang *et al.*, “A functionalized ionic liquid as the next-generation nano-lubricant,” *Droplet*, vol. 1, no. 2, pp. 192–201, Oct. 2022, doi: 10.1002/dro2.28.
- [5] L. Landau and B. Levich, “Dragging of a liquid by a moving plate,” *Acta Physicochim. URSS*, vol. XVII, 1942.